

SQL Fundamentals

Updated Exercises

Exercise 5: NULL Functions

Table 1: Employees

☛ Show all employees with their salary. If salary is NULL, display 0.

Q1

```
SELECT employee_id,  
       name,  
       COALESCE(salary, 0) AS salary_with_default  
FROM Employees;
```



employee_id	name	salary_with_default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2

```
SELECT employee_id,  
       name,  
       COALESCE(department, 'Not Assigned') AS department_name  
FROM Employees;
```



employee_id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Table 2 : Orders

Q3 : Find orders with NULL customer_id using ISNULL()

```
SELECT order_id,  
       customer_id  
FROM Orders  
WHERE customer_id IS NULL;
```

↓

order_id	customer_id
103	NULL

Q4: Show all orders. If delivery_date is NULL, show 'Pending'

```
SELECT order_id,  
       customer_id,  
       IFNULL(delivery_date, 'Pending') AS delivery_status  
FROM Orders;
```

↓

order_id	customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Table 3 : Students

```
Q5. SELECT student_id,  
         name,  
         IFNULL(grade, 0) AS final_grade  
FROM Students;
```

↓

student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Q6.
 SELECT
 COUNT(*) AS not_graded_count
 FROM Students
 WHERE final_grade = 0;

↓

not_graded_count
1

Table 4 : Products

Q7.
 SELECT product_id,
 name,
 price - IFNULL(discount, 0) AS final_price
 FROM Products;

↓

product_id	name	final_price
501	Keyboard	25
502	Mouse	10
503	Monitor	100

Table 5 : Customers

Q8. SELECT
COUNT(*) AS missing_email_count
FROM Customers
WHERE email IS NULL;

↓

missing_email_count
2

Q9. SELECT customer_id,
name,
COALESCE(email, 'No Email') AS email_display
FROM Customers;

↓

customer_id	name	email_display
1	Linde	No Email
2	Joseph	joseph@mail.com
3	Nia	No Email

Table 6 : Payments

Q10. SELECT payment_id,
IFNULL(method, 'Unknown') AS method_display,
status
FROM Payments;

↓

payment_id	method_display	status
301	Credit	NULL
302	PayPal	Success
303	Unknown	Failed

Table 7: Inventory

Q₁₁. SELECT item_id,
item_name,
IFNULL(quantity, 0) AS quantity_checked
FROM Inventory;



item id	item name	quantity_checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Table 8: Employees_Extra

Q₁₂. SELECT emp_id,
COALESCE(bonus, commission) AS first_available_reward
FROM Employees_Extra;



emp_id	first_available_reward
1	300
2	100
3	NULL

Table 9: Classes

Q13. SELECT
COUNT(*) AS no_room_count
FROM Classes
WHERE room IS NULL;

↓

no_room_count
2

Table 10: Attendance

Q14. SELECT student_id,
date,
IFNULL(status, 'Not Marked') AS attendance_status
FROM Attendance;

↓

student_id	date	attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

Table 11: Bank Accounts

Q15. SELECT account_id,
IFNULL(account_type, 'Unknown') AS type_display,
IFNULL(balance, 0) AS balance_checked
FROM Bank_Accounts;

↓

account_id	type_display	balance_checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Table 12 : Projects

Q16 : `SELECT project_id,
title,
IFNULL(start_date, 'TBD') AS start_display
FROM Projects;`

project_id	title	start_display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Table 13 : Reviews

Q17 `SELECT review_id,
product_id,
IFNULL(comment, 'No Comment') AS comment_display,
IFNULL(rating, 0) AS rating_display
FROM Reviews;`

review_id	product_id	comment_display	rating_display
1	501	Great product	4
2	502	No Comment	0
3	503	Works fine	3

Table 14 : Suppliers

Q18. SELECT supplier_id,
name,
COALESCE (phone, alt_phone, 'No Contact') AS
contact_number
FROM Suppliers ;

supplier_id	name	contact_number
1	Global Goods	123 456 789
2	Best Supplies	987 654 321
3	ValueScore	No Contact

Table 15 : User_Settings

Q19. SELECT user_id,
IFNULL (theme, 'Light') AS theme_set,
IFNULL (language, 'English') AS language_set,
IFNULL (timezone, 'UTC') AS timezone_set
FROM User_settings ;

user_id	theme_set	language_set	timezone_set
1	Light	English	UTC
2	Dark	English	UTC+1
3	Light	English	UTC

Table 16: Maintenance

Q 20. `SELECT record_id,
machine_id,
IFNULL(issue, 'Unknown Issue') AS issue_log,
IFNULL(technician, 'Not Assigned') AS technician_name
FROM Maintenance;`

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	Jammed	Alex